

THE MINISTRY OF SCIENCE AND HIGHER EDUCATION
OF THE RUSSIAN FEDERATION



ITMO UNIVERSITY

International Research and Educational Center
for Physics of Nanostructures

LAB 1
for the Course
"Adaptive and Robust Control"

on the topic:
SIMPLE EXAMPLE OF ADAPTIVE CONTROLLER DESIGN

Date:
Group No. 20

October 22, 2025
Variant: 15

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INTRODUCTION

0.1 Introduction

Problem statement. Consider the scalar plant

$$\dot{x} = \theta x + u,$$

where x is the state variable (coincides with the output), u is the control, θ is the unknown scalar parameter.

The objective is to design a control that will compensate the influence of θ on the closed-loop system stability and ensure the limiting equality

$$\lim_{t \rightarrow \infty} (x_m(t) - x(t)) = \lim_{t \rightarrow \infty} \varepsilon(t) = 0,$$

where $\varepsilon = x_m - x$ is the control error, x_m is the output of the reference model

$$\dot{x}_m = -\lambda x_m + \lambda g,$$

with a piece-wise continuous and bounded reference signal g and the parameter $\lambda > 0$ responsible for the transient performance of closed-loop system after adaptation process.

Problem solution. Solution consists of the adjustable control

$$u = -\hat{\theta}x - \lambda x + \lambda g.$$

and the adaptation algorithm

$$\dot{\hat{\theta}} = -\gamma x \varepsilon$$

with adaptation gain $\gamma > 0$.

0.2 The order of the work

1. Using the parameters of the nonlinear modification of the plant given by

$$\dot{x} = \theta f(x) + u,$$

where $f(x)$ is a continuous function, and the reference given in Table 1 design the corresponding nonadaptive controller.

Make a simulation experiment, in which the plant parameter θ is increased rapidly (controller parameters are not changed) so that the closed-loop system loses stability. Plot separately the variable x together with x_m and variable u .

2. Design the adaptive control and repeat Experiment №1 with changing plant parameters and $\hat{\theta}(0) = 0$. Adaptation gain γ is selected experimentally based on better representation of results. Plot separately the variable x together with x_m , variable u and additionally variable $\tilde{\theta} = \theta - \hat{\theta}$.

3. Make experiment for different gains γ .

0.3 Notations

Throughout this work, the following notations are used:

Table 1 — Table 1: Notations of symbols

Symbol	Explanation
x	State variable
u	Control
θ	Scalar parameter
ε	Control error
x_m	Output of the reference model
g	Reference signal
λ	Parameter for transient performance

0.4 Variant

Table 2 — Objective function and constraint

Var.	θ	$f(x)$	λ	$g(t)$
15	29°	x^4	6	$\cos(2t) \sin(t) + 1$

1 SOLUTIONS

1.1 Design the non-adaptive controller

The plant is given by the equation:

$$\dot{x} = \theta f(x) + u \quad (1)$$

where $f(x)$ is a continuous function, and the reference given in Table 1 design the corresponding nonadaptive controller.

The reference model is given by:

$$\dot{x}_m = -\lambda x_m + \lambda g \quad (2)$$

The objective is to design a control that will compensate the influence of θ on the closed-loop system stability and ensure the limiting equality

$$\lim_{t \rightarrow \infty} (x_m(t) - x(t)) = \lim_{t \rightarrow \infty} \varepsilon(t) = 0 \quad (3)$$

where $\varepsilon = x_m - x$ is the control error, x_m is the output of the reference model with a piece-wise continuous and bounded reference signal g and the parameter $\lambda > 0$ responsible for the transient performance of closed-loop system after adaptation process.

The error dynamics is given by:

$$\dot{\varepsilon} = \dot{x}_m - \dot{x} \quad (4)$$

Substituting the equations of the plant and the reference model into the error dynamics, we get:

$$\dot{\varepsilon} = -\lambda x_m + \lambda g - \theta f(x) - u \quad (5)$$

The solution consists of the adjustable control

$$u = -\theta f(x) - \lambda x + \lambda g \quad (6)$$

Make a simulation experiment, in which the plant parameter θ is increased rapidly (controller parameters are not changed) so that the closed-loop system loses stability.

1.2 Simulink Model

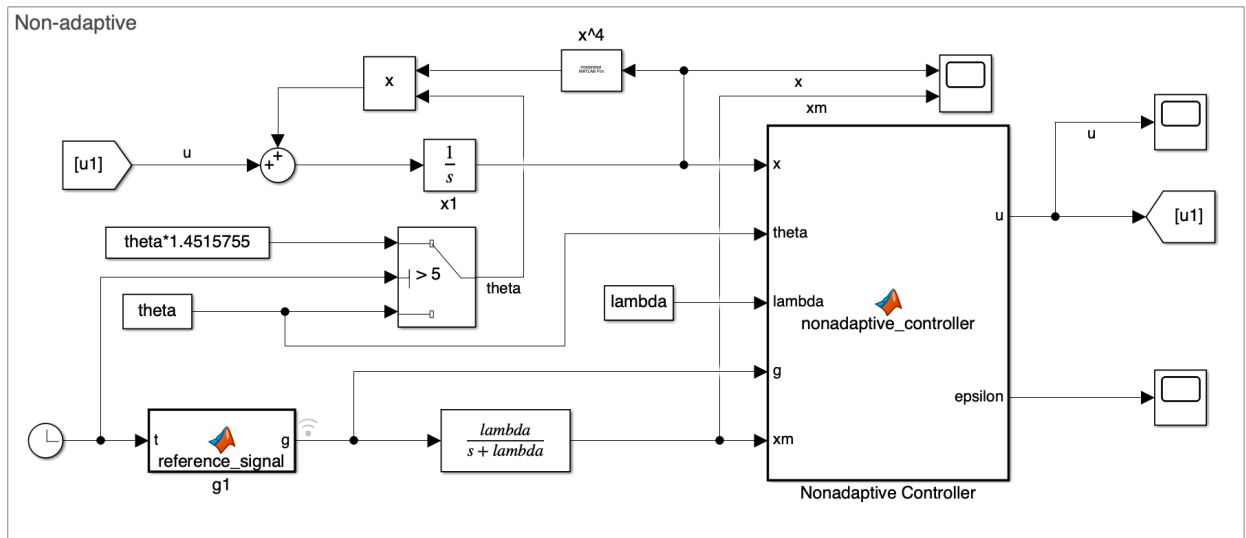
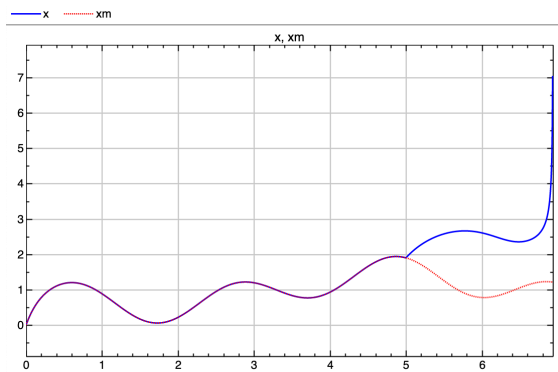
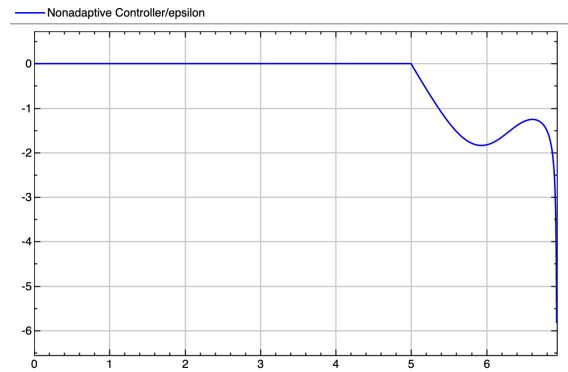


Figure 1 — Non-adaptive Controller

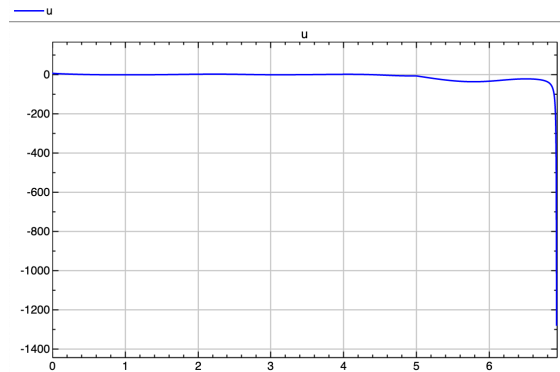
Theta will be increased at time $t=5s$.



(a) Xm-X Output



(b) Error



(c) Control Input

Figure 2 — Simulation Results of Non-adaptive Controller

The plots show the system becoming unstable after theta is increased at $t=5s$.

1.3 Design the adaptive controller

Using the same plant and reference model as in Section 1, design an adaptive controller that will compensate the influence of the unknown parameter θ on the closed-loop system stability and ensure the limiting equality

$$\lim_{t \rightarrow \infty} (x_m(t) - x(t)) = \lim_{t \rightarrow \infty} \varepsilon(t) = 0 \quad (7)$$

where $\varepsilon = x_m - x$ is the control error, x_m is the output of the reference model with a piece-wise continuous and bounded reference signal g and the parameter $\lambda > 0$ responsible for the transient performance of closed-loop system after adaptation process. The error dynamics is given by:

$$\dot{\varepsilon} = \dot{x}_m - \dot{x} \quad (8)$$

Substituting the equations of the plant and the reference model into the error dynamics, we get:

$$\dot{\varepsilon} = -\lambda x_m + \lambda g - \theta f(x) - u \quad (9)$$

The solution consists of the adjustable control

$$u = -\hat{\theta} f(x) - \lambda x + \lambda g \quad (10)$$

and the adaptation algorithm

$$\dot{\hat{\theta}} = -\gamma f(x) \varepsilon \quad (11)$$

where $\hat{\theta}$ is the estimate of the unknown parameter θ , and $\gamma > 0$ is the adaptation gain.

1.4 Simulink Model

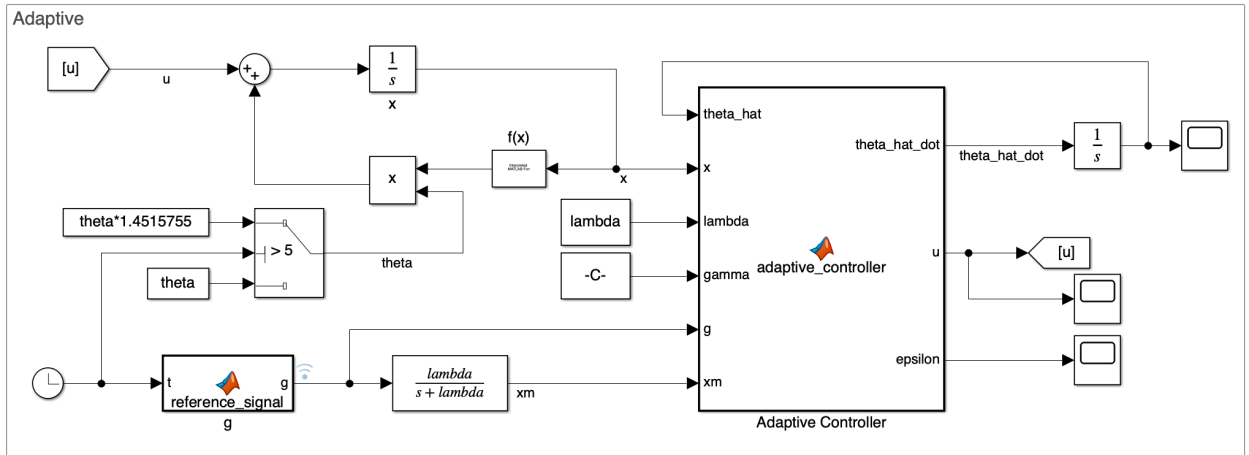
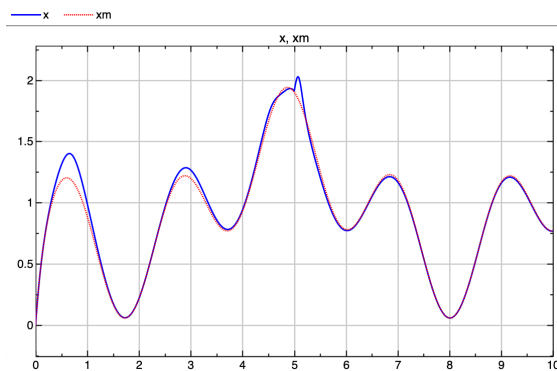
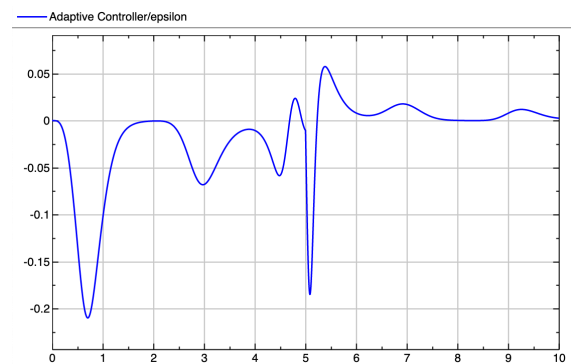


Figure 3 — Adaptive Controller

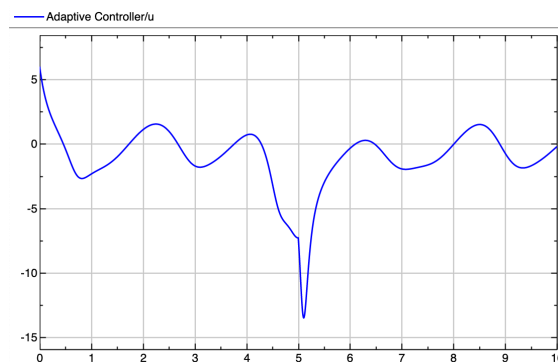
Theta will be increased at time $t=5s$. And $\gamma = 1$.



(a) Xm-X Output



(b) Error



(c) Control Input

Figure 4 — Simulation Results of Adaptive Controller

The plots show the system remaining stable after theta is increased at $t=5s$.

1.5 Varying the adaptation gain γ

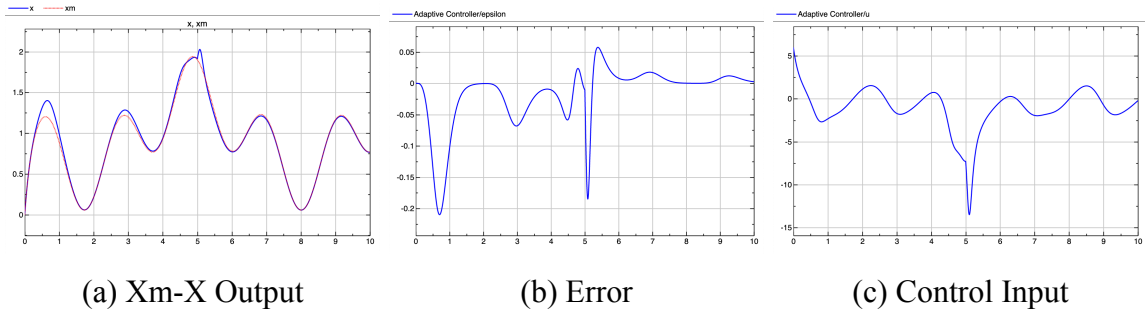


Figure 5 — $\gamma = 1$

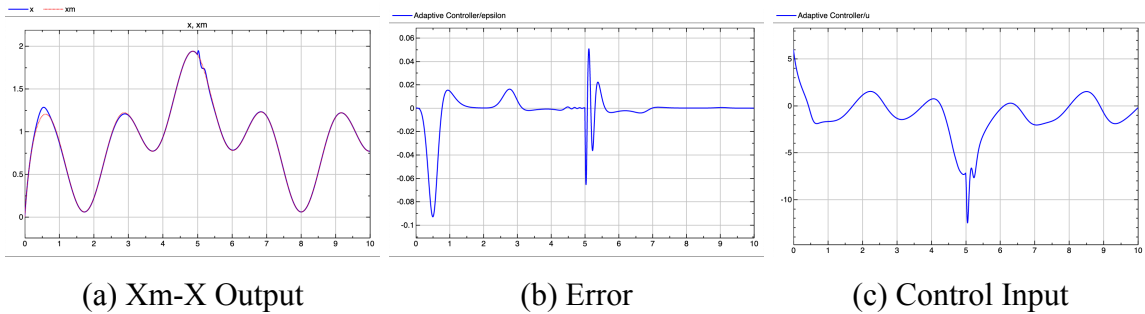


Figure 6 — $\gamma = 10$

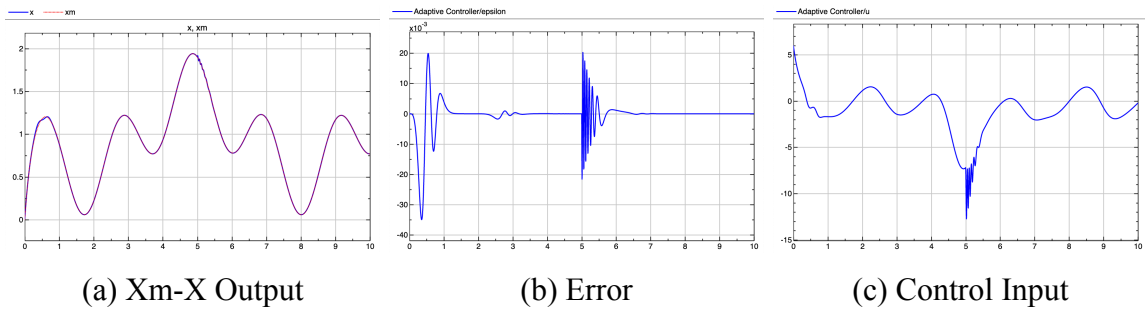


Figure 7 — $\gamma = 100$

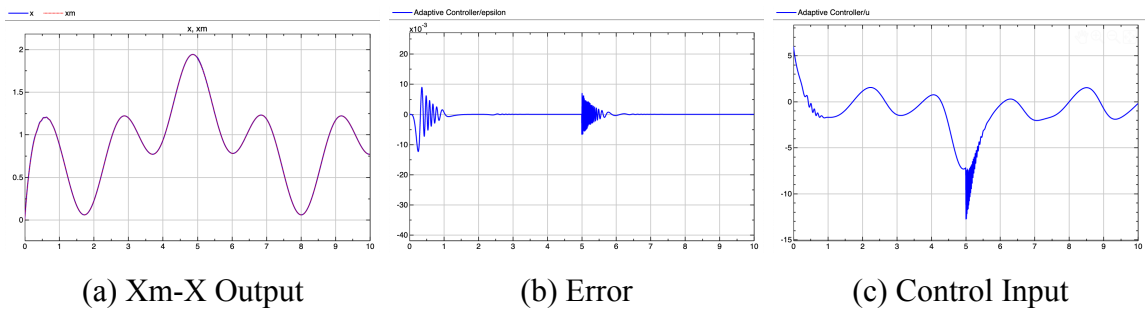


Figure 8 — $\gamma = 1000$

2 CONCLUSION

In this lab work, I designed and simulated both non-adaptive and adaptive controllers for a scalar plant with an unknown parameter. The non-adaptive controller demonstrated instability when the plant parameter was rapidly increased, highlighting its limitations in handling parameter variations. In contrast, the adaptive controller successfully compensated for the changes in the plant parameter, maintaining system stability and ensuring that the control error converged to zero over time.

The results underscore the importance of adaptive control strategies in systems with uncertain or varying parameters, providing a robust solution to maintain desired performance levels.

And when γ increased from 1 to 1000, the adaptive controller responded faster to changes in the plant parameter, further enhancing system stability and performance. But the frequency of oscillations also increased, indicating a trade-off between response speed and system stability.

This demonstrates the effectiveness of tuning adaptation gains in adaptive control systems to achieve optimal response characteristics.